



MIRA

robotic camera-inspection assistant

An arm that inspects the physical world — steered from a phone in XR, fully offline.

TEAM JQK

QUALCOMM HACK CHALLENGE

FINALS • 2026 • SHEET 1 OF 1

PLATFORM: ARDUINO UNO Q

1 • THE PRODUCT

An SO-101 robot arm carries a camera. You steer it from a phone in XR; it scans, inspects, records and replays trajectories over real objects.

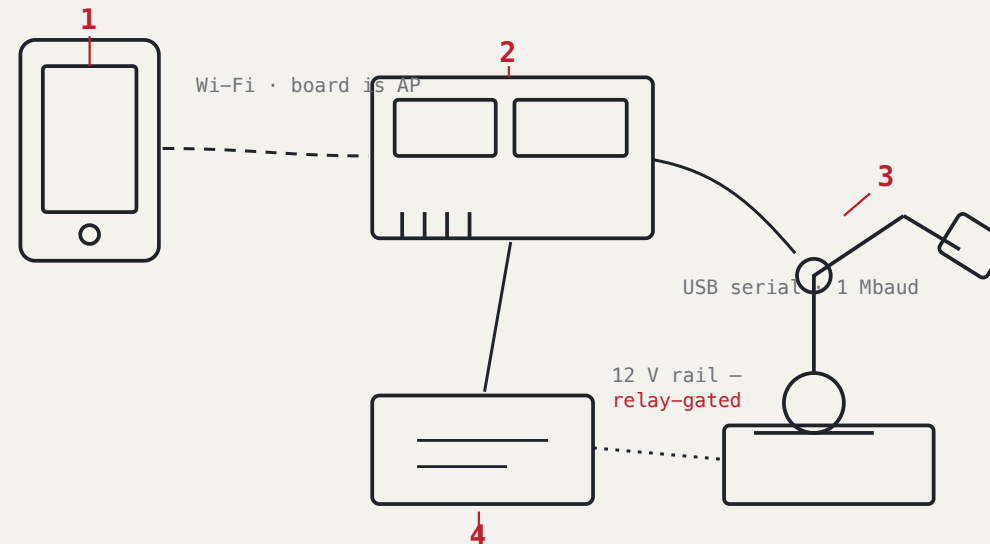
2 • THE POINT

Close-up inspection with no cloud, no GPU, no connectivity — every frame stays on the 2 GB board. *Private by physics, not by policy.*

PHOTOGRAPH № 1

team at the rig — to be affixed
(awaiting asset from team)

FIG. 1 — SIGNAL PATH



1 phone — WebXR teleop UI

2 UNO Q — capture • control • server

3 SO-101 arm • wrist camera

4 STM32 interlock — gates 12 V rail

3 • STATUS LEDGER

100 Hz servo control	✓ working
Camera capture	✓ working
Trajectory replay	✓ working
Servo-power interlock	✓ working
XR teleop (WebXR)	◦ in progress
3D reconstruction	• planned
Voice control	• planned

2 GB RAM TOTAL	0 GPU • CLOUD	100% ON-DEVICE
6 SERVOS	2 CAMERAS	1 RELAY

NOTICE — SAFETY IS HARDWARE

Cutting board power does **not** stop the arm; servos freeze rigid. An STM32 relay gates the 12 V servo rail — Linux cannot override it.
heartbeat stale → **rail drops** • E-stop opens → **rail drops** • corrupt frame → **rail drops** • faults latch until a human resets

